

## MASTER'S COMPREHENSIVE EXAMINATION

### Part II Applied (2 hours) March 30<sup>th</sup>, 2019

**Instructions:** Attempt any three out of the following four questions on all three pages. Indicate which three you would like to have graded. Open book and notes, plus non-programmable calculators can be used. No computers, programmable calculators, cell phones or other communications / web accessing devices.

#### Put Your Name Both on the Blue Book and the Test Sheet

1. Captain Krik has obtained 5 independent random observations on the magnitude of interstellar blips which have the following values; -7, -4, -1, 4, 8. These are presented as single digits to simplify computation, but assume for your answers these are taken down to two decimal places.
  - a. Give the best estimates from this sample for ;the population mean, standard deviation, skewness and kurtosis. Show your formulas and calculations.
  - b. Assume that these five values come from a normal distribution and that the next interstellar blip to occur will be random / independent. Give a 95% confidence interval for the value of this next interstellar blip out to 2 decimal places
2. The shelf lives of two types of carbonated beverages A and B are to be compared. It is only important to know if the average shelf life of beverage B is longer than that of the average shelf life of beverage A. Ten bottles of A and 12 bottles of B are tested for shelf lives and the following data are obtained:  
  
Beverage A    119, 127, 117, 141, 128, 117, 130, 132, 131, 122  
Beverage B    126, 134, 124, 148, 135, 124, 137, 139, 138, 129, 147, 136
  - a) State the null hypothesis that you would test. Also state the alternative hypothesis.
  - b) Assuming that the two groups of observations come from populations with the same variance, suggest an appropriate test statistic and calculate its value. State any other assumptions that you make.
  - c) Let  $pt(q,v)$  denote the area below  $q$  in the curve of the  $t$ -distribution with  $v$  degrees of freedom. Given the following:  $pt(2.482,10)= 0.9837842$ ,  $pt(2.482,20)= 0.9889678$ ,  $pt(2.482,30)= 0.9905558$ , what is the  $p$ -value associated with the computed value of the test statistic? Can you reject the null hypothesis at the 5% level of significance? Can you reject it at the 1% level?
  - d) Obtain a 95% confidence interval for the difference between the average shelf lives of beverage B and beverage A. Use the information from the following that you need for this purpose:  $pt(2.84534,20)=.995$ ,  $pt(2.527977,20)= .99$ ,  $pt(2.085963,20) = .975$ ,  $pt(1.724718,20)=.950$ .

3. A textile company conducted an experiment to study whether K machines were able to provide similar measurements of fiber strength. From the same batch, K x M different samples of fiber were taken, and randomly assigned to the K machines so that each machine measured M similar fiber samples. A one-way analysis of variance (ANOVA) was performed with the experimental data. Answer the following questions.

➤ anova(a)

➤

Response: y (measured fiber strength)

|           | Df | Sum Sq  | Mean Sq | F value |
|-----------|----|---------|---------|---------|
| Machine   | 2  | 11.1049 | ?       | ?       |
| Residuals | 12 | ?       | 0.3541  |         |

- How many machines were compared in the study, i.e., what is the value of K?
  - How many samples of fiber did each machine measure, i.e., what is the value of M?
  - Write down the statistical model for the above analysis of variance.
  - Fill in the missing values in the ANOVA table. Write this in your blue book not on the test sheet.
  - Provide an estimate of the residual standard error.
  - Let  $pf(q,a,b)$  denote the area below q in the curve of the F-distribution with a,b degrees of freedom. If  $1-pf(10,2,12)=.003$ , what can you conclude from the ANOVA table?
4. Dr. AP has used independent observations from 100 federally funded projects to fit a linear regression model to  $Y = \text{Deficit Spending}$  based on  $X_R = \text{Resources Score}$  and  $X_N = \text{Needs Score}$ . The model fit is

$$Y = a + b_R X_R + b_N X_N + \varepsilon,$$

where “Resources\_Score” and “Needs\_Score” are continuous variables taking on both positive and negative values and  $\varepsilon$  is normally distributed random error with mean 0 and constant variance. Note that a positive value of Y means that the project will have a spending deficit, a negative value means the project will have a spending surplus, and a value of zero means the project will break even.

Dr. AP obtains the following parameter estimates:

| Predictor       | Point Estimate  | Std-Err |
|-----------------|-----------------|---------|
| Intercept       | $\hat{a}$ 10.5  | 5.2     |
| Resources_Score | -5.6            | 1.8     |
| Needs_Score     | $\hat{b}_N$ 7.8 | 3.0,    |

with the following covariances  $Cov(\hat{a}, \hat{b}_R) = 2.5$ ,  $Cov(\hat{a}, \hat{b}_N) = -3.2$ ,

$$Cov(\hat{b}_R, \hat{b}_N) = -1.0.$$

- A. Test the null hypothesis that a project with both Resources\_Score ( $X_R$ ) and Needs\_Score ( $X_N$ ) equaling 0 will have an expected deficit spending of 0 against a two sided alternative at  $p = 0.01$
- B. Give a 95% confidence interval for the expected deficit spending of a project with a Resources\_Score  $X_R = 1$  and Needs\_Score  $X_N = 1$ .
- C. Assume there are two Projects: A. Project-I has a Resources\_Score  $X_R = a$  and Needs\_Score  $X_N = b$  B. Project-II has a Resources\_Score  $X_R = a+1$  and Needs\_Score  $X_N = b+1$  for the same values  $a$  and  $b$ . Calculate the p-value testing the null hypothesis that Project-II has the same expected deficit spending than does Project-I, vs. the alternative that Project-II has a higher expected deficit spending.